Impact of Digital Government Construction on Green Total Factor Productivity: A Quasi-natural Experiment with Double Debiased Machine Learning

Abstract

The deep-seated contradiction between the green development of Chinese cities and the barriers to government information has transformed the digital governance paradigm centered on constructing a digital government. This study takes 278 cities in China as the experimental objects and takes the policy of big data management institution reform as a quasi-natural experiment. By constructing the DDML-DID model, this study examines the impact of digital government construction on green total factor productivity (GTFP). Various machine learning algorithms indicate that for every one-unit increase in the digital government construction, the urban GTFP can be significantly enhanced by 0.9% to 4.7%, and there are varying degrees of individual policy responses within the range of -0.1 to 0.2. Mechanism analysis indicates that the digital government construction encourages the improvement of GTFP through a dynamic chain transmission path with varying policy sensitivities, such as enhancing governmental digital attention, improving digital infrastructure construction, promoting digital technological innovation, and promoting digital economic development. Heterogeneity analysis indicates that the promoting effect of digital government construction on GTFP is more significant in ecological function areas, resource-based cities, non-two-control areas and central cities. The research conclusion of this study provides valuable inspiration for the government to break through the institutional roots that hinder the urban green development through digital transformation.

Keywords Digital government; Government construction; Big data management; Green total factor productivity; Green development

1 Introduction

Against the backdrop of mounting global climate governance pressures and an accelerated low-carbon transition, green total factor productivity (GTFP) has emerged as a core indicator measuring the quality of urban green transformation and a key engine for achieving China's "dual carbon" goals. Improving GTFP not only contributes directly to Sustainable Development Goals (SDGs)—particularly SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action)—but also embodies a critical pathway for synergizing economic growth with environmental sustainability. However, systemic inefficiencies in governmental governance, especially the persistent "data island" phenomenon within traditional administrative systems, have led to severe fragmentation of environmental information. The lack of cross-departmental collaboration weakens environmental regulation enforcement, impedes the transmission of emission reduction policies, and dampens incentives for green development. These governance shortcomings form a structural bottleneck constraining the improvement of GTFP.

The global digital revolution offers a historic opportunity to reshape public governance paradigms. Cutting-edge digital technologies—such as big data, artificial intelligence, and blockchain—are driving governmental capacity toward digitalization and intelligence. For example, cross-tier data sharing platforms can integrate information flows across environmental monitoring, energy consumption, and industrial operations. Such digital empowerment not only mitigissues of governance fragmentation caused by "data islands" but also enhances the precision of environmental regulation and market responsiveness, thereby injecting new momentum into GTFP growth.

To address these issues, this study examines 278 prefecture-level cities in China from 2007 to 2023, leveraging the reform of big data management institutions as a quasi-natural experiment. By constructing a DDML-DID model that integrates double debiased machine learning with staggered DID, this study empirically investigate the impact of digital government construction on GTFP. This study aims to achieve the following research objectives: First, to accurately identify the net effect of digital government construction on urban GTFP; Second, to unpack the underlying mechanisms—including governmental digital attention, digital infrastructure, technological innovation, and digital economy development;

Third, to explore heterogeneous policy effects across different types of cities.

This study offers several marginal contributions: First, this study incorporates the "information-power" theory from political science into a green productivity framework, constructing an interdisciplinary transmission logic of "breaking data islands – reconstructing information power – enhancing green productivity," addressing the oversight of institutional roots in existing technical-oriented studies; Second, this study employs a DDML-DID model to overcome issues of non-random policy adoption, nonlinear relationships, and high-dimensional control variables, providing robust causal identification; Third, this study proposes a dynamic chain-mediated mechanism spanning institutional, infrastructural, technological, and economic dimensions, revealing the logical nestedness and evolutionary path of digital government's impact; Forth, this study integrate urban ecological functions, resource endowments, environmental goals, and city tiers into a holistic analytical framework, identifying heterogeneous impacts and providing differentiated policy insights for advancing GTFP.

2 Policy review and literature review

2.1 Policy review

From the perspective of political science, data information is a crucial political resource and a source of political power in the new era (Dahl, 2007). Due to the deep-seated connection between data information and political power, there is no inherent incentive to fully share data information among bureaucratic organizations, which creates the "data island" that hinders the exchange of data information among governments and enterprises. In reality, China's data system and administrative system have lagged behind social and economic development for a long time. Due to the data monopoly of various departments, data fragmentation is formed, and key green data information related to environmental protection and energy is deeply trapped in the "power island", which seriously hinders the value production of data elements. Therefore, data deregulation through data management system reform is the key to activating the data's potential and a necessary measure to improve GTFP.

In response to the global digital revolution, in August 2015, the General Office of the State Council of China issued the "Action Outline for Promoting the Development of Big Data", encouraging governments at all levels to explore and establish big data institutions. In March

2018, the Central Committee of the Communist Party of China issued the "Plan for Deepening the Reform of Party and State Institutions", explicitly stating that local authorities can set up institutions and allocate functions within the prescribed limits by local conditions. Under the guidance of this document, provincial governments have initiated the reform of government institutions, among which the big data management institution reform has become one of the biggest highlights of this institutional reform. In October 2023, the National Data Bureau was officially established, marking the basic completion of the "bottom-up" data system reform from the central government to local governments in China, and is expected to solve problems such as "data island" at the national level.

As of December 2024, all 31 provincial administrative regions in China (excluding Hong Kong, Macao and Taiwan) have reformed big data management institutions and established data management institutions at different levels, except for the Ningxia Hui Autonomous Region. The policy review figure of the big data management institution reform in various provinces of China is shown in Fig 1.

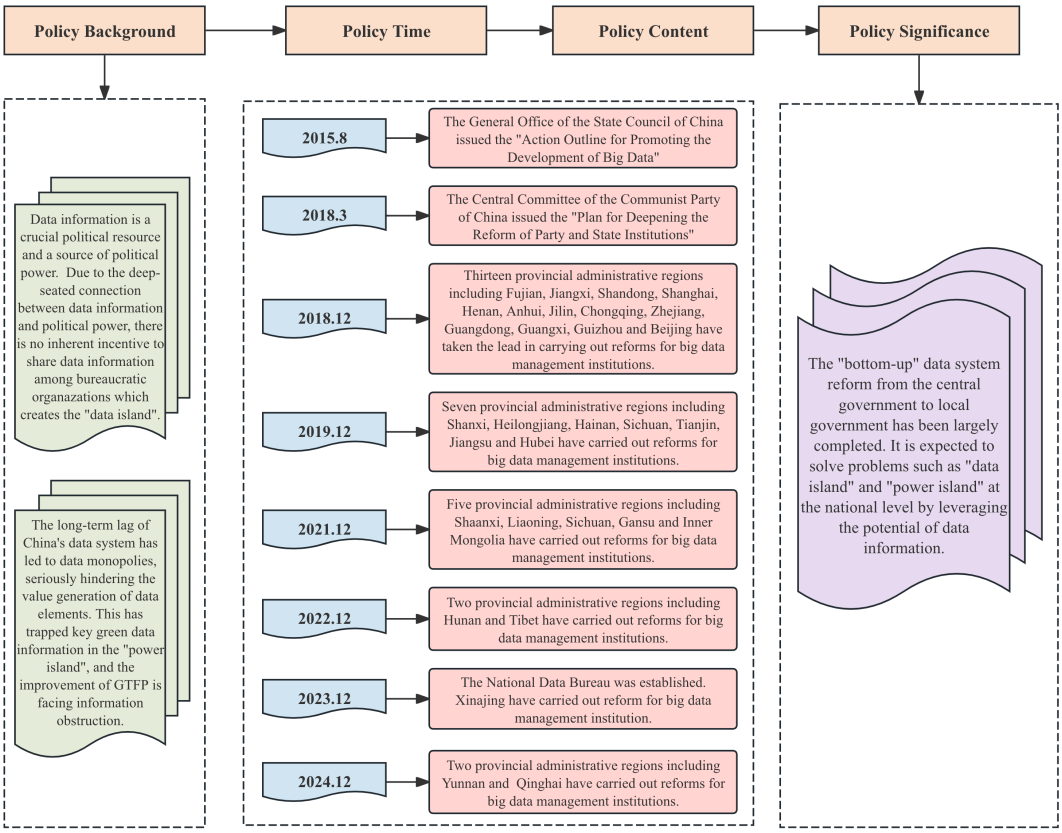

Fig. 1 The policy review of big data management reform

2.2 Literature review

2.2.1 The research on GTFP

Existing research on GTFP primarily focuses on identifying its key determinants and operational mechanisms. First, digitalization and technological innovation are regarded as crucial drivers for enhancing GTFP. ICT development not only directly promotes green total factor energy productivity but also generates moderating effects through technological innovation and industrial structure upgrading. Its impact exhibits nonlinear characteristics with varying levels of resource allocation (Gao et al., 2022). Similarly, digital economic development exhibits a U-shaped direct effect and spatial spillover effects on GTFP, with its channels including driving green technological progress, promoting industrial structure upgrading, and alleviating factor market distortions (Lyu et al., 2023).

Second, green finance and financial development have also been proven to significantly promote GTFP, with more pronounced effects in regions with stronger economic foundations or more severe environmental pollution (Lee and Lee, 2022; Jiakui et al., 2023). Notably, financial agglomeration significantly boosts local GTFP but inhibits GTFP in surrounding cities (Xie et al., 2021). Third, environmental regulations exhibit marked heterogeneity in their impact on GTFP, with effects contingent on regional political attributes and development stages. In regions with high political attributes, local environmental regulations exert positive impacts on GTFP and generate spatial spillovers, whereas in regions with low political attributes, they may induce negative effects due to "race-to-the-bottom competition" (Li and Wu, 2017).

Furthermore, factor market distortions have been shown not only to suppress GTFP growth but also to weaken the positive spillovers of exports and FDI (Lin and Chen, 2018). Finally, other factors such as fiscal decentralization (Song et al., 2018), human capital structure (Wang et al., 2021), internet development (Wu et al., 2021), and green innovation (Wu et al., 2022; Zhao et al., 2022) have been shown to influence GTFP through pathways such as technological progress, resource allocation, or institutional environments under different contexts.

2.2.2 The research on digital government

In the field of digital government research, existing research has explored multiple dimensions including conceptual evolution, institutional effectiveness, and corporate responses. First, the evolution of digital government has been summarized as progressing from a technological phase, through e-government, to a policy-driven, contextualized stage of e-governance. This trajectory reflects its increasingly complex and specialized developmental path (Janowski, 2015). During this process, the concept of “digital government inclusivity” has gained prominence, emphasizing that automated government procedures should balance fairness and participation while proposing mitigation mechanisms to foster inclusivity (Peeters et al., 2025).

Second, digital government development significantly enhances administrative efficiency, primarily by optimizing internal functions and overall operational effectiveness through coordinated digital transformation (Yang et al., 2024). Additionally, at the platform governance level, designing reasonable boundary resources facilitates coordinated control and collaboration, driving cross-agency service innovation (Gong and Li, 2023).

Third, digital government profoundly influences corporate behavior. By leveraging data openness, cloud platforms, and intelligent services, it reduces institutional transaction costs for enterprises, thereby enhancing their ESG performance and fostering technological innovation and intelligent transformation (Chen et al., 2025; Zhang and Zhang, 2025). Moreover, digital government enhances regulatory capabilities, improves R&D subsidy efficiency and corporate innovation quality (Wang et al., 2025), and stimulates corporate investment by reducing policy uncertainty, optimizing resource allocation, and strengthening credit supply (Huang et al., 2025).

2.2.3 The research on digital government, environment and energy

In cross-disciplinary research on digital government within environmental and energy sectors, existing studies primarily focus on its role in advancing green development and sustainable energy. From the energy perspective, digital government significantly promotes energy sustainability through mechanisms such as enhancing energy efficiency, stimulating innovation, and driving regional digitalization. These effects exhibit heterogeneity across cities with varying resource endowments and geographical locations (Jiang et al., 2024; Jiang, 2025). Regarding environmental impacts, digital government exhibits an inverted U-shaped relationship with carbon emissions, initially promoting increases before later suppressing them. Once development surpass a specific threshold, it generates carbon reduction effects by enhancing administrative efficiency, regulatory quality, and the business environment (Li et al., 2025). Similarly, government digitalization can indirectly reduce carbon emissions by fostering digital economic growth (Zhang et al., 2025), while digital government has been proven to curb emissions through industrial structure upgrading and green technological innovation (Yi, 2025).

From the macro perspective, digital government positively alleviates energy poverty through primarily technological and informational effects, with the magnitude of these effects moderated by the severity of energy poverty and country type (Feng et al., 2025). From the micro perspective, digital government enhances firms' GTFP through two pathways: resource acquisition and resource restructuring. Specifically, it alleviates financing constraints and strengthens green innovation capabilities (Jiang, 2025). Digital government also improves energy efficiency in resource-based city enterprises by driving technological progress and reducing non-productive expenditures (Tang et al., 2025). Furthermore, digital government promotes corporate green innovation by optimizing the business environment, reducing transaction costs, and increasing R&D investment (Tan et al., 2024), thereby reinforcing its enabling role in environmental governance and green transition.

The existing literature provides valuable theoretical foundations and empirical evidence for understanding the complex relationship between digital government and GTFP. However, there remains considerable scope for deepening and expanding current research. First, most studies focus on examining the impact of digital technologies or the digital economy on GTFP from a singular technological or economic perspective, or analyze the isolated effects of digital government on corporate behavior or specific environmental factors. This approach tends to overlook the core mechanism whereby digital government itself—as a systemic institutional arrangement—can drive GTFP growth by restructuring information power dynamics within government. Second, methodologically, traditional policy evaluation models may face challenges of selection bias and model specification errors when applied to research contexts like digital government, which may involve non-random pilot selection and complex nonlinear relationships among high-dimensional variables. Building upon existing research contributions, this study aims to conduct in-depth exploration and supplementation addressing these limitations.

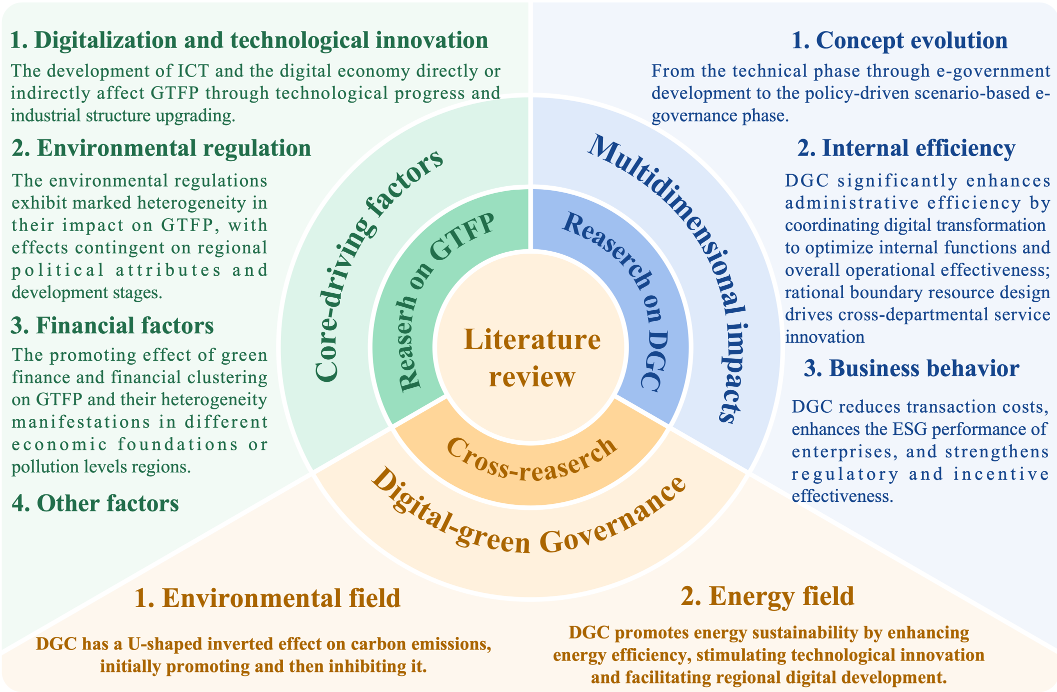

Fig. 2 Literature review circular chart

3 Theoretical analysis

3.1 Direct effect of digital government construction on GTFP

As a paradigmatic embodiment of digital government construction, the reform of big data management institutions fundamentally dismantles the "data island" phenomenon by reconstructing the information power structure within hierarchical organizations (Wilson and Mergel, 2022). Rooted in the "information-power" theory of political science, this institutional transformation addresses the fragmented division of the traditional administrative system that had led to departmental monopolization of environmental data, creating substantial information barriers for green governance—manifested through the dispersion of critical data on energy consumption and pollution emissions. The establishment of dedicated government big data management institutions (e.g., the National Data Bureau and local data bureaus) systematically breaks departmental "data sovereignty" through institutional mechanisms, converting previously isolated environmental data into public governance resources via mandatory collection of cross-departmental ecological information and the creation of unified sharing platforms.

This redistribution of informational power translates into measurable improvements in green total factor productivity (GTFP) through two interconnected pathways. Firstly, the enhanced information accessibility and quality enable precise identification of resource misallocation patterns and pollution sources, allowing governments to dynamically optimize environmental policy tools and improve resource allocation efficiency. By reducing information asymmetry in environmental regulations, authorities can implement targeted interventions that correct market failures and reduce negative environmental externalities Secondly, the integration of previously fragmented environmental data facilitates smarter regulatory oversight and enforcement, enhancing the effectiveness of environmental supervision mechanisms. The centralized data platform enables real-time monitoring of pollution activities, predictive analysis of environmental risks, and more efficient deployment of regulatory resources, thereby simultaneously promoting production efficiency while reducing ecological impacts.

Based on the above theoretical analysis, this study constructs a mathematical model to further explore the impact of digital government construction on GTFP:

First, based on classical production theory and endogenous technological progress models, this study assumes a Cobb-Douglas production function for consensual output and introduces the information factor to influence technological efficiency:

$$Y_{it}=A(I_{it})\times K_{it}^{\alpha}L_{it}^{\beta}E_{it}^{\gamma} \tag{1}$$

where Y_{it} represents the desirable output (urban GDP); K_{it}^{α} represents capital input; L_{it}^{β} represents labor input; E_{it}^{γ} represents energy input; $A(I_{it})$ represents technical efficiency, influenced by information factors; α, β, γ represent output elasticities and $\alpha+\beta+\gamma=1$ (constant returns to scale).

Second, based on the emission generation function in environmental economics, this study constructs an urban pollution function:

$$P_{it}=k\times Y_{it}^{\phi}E_{it}^{\psi}\times\vartheta(I_{it})^{-\eta} \tag{2}$$

where P_{it} represents the undesirable output (sulfur dioxide, industrial smoke and industrial wastewater); $\vartheta(I_{it})$ represents environmental governance efficiency, influenced by information factors I_{it}; k, φ, ψ, η represent positive parameter.

Third, the construction of the GTFP function must simultaneously account for both desirable and undesirable outputs. To facilitate model derivation, this study employs an exponential approximation for the directional distance function:

$$GTFP_{it}=A(I_{it})\times exp(-\delta P_{it}) \tag{3}$$

where $\delta>0$ represents the pollution loss coefficient, reflecting the negative impact of pollution on overall efficiency.

Fourth, based on the theory of diminishing marginal returns to information in information economics, this study assume that technical efficiency and environmental governance efficiency are power functions of the information factor:

$$A(I_{it})=A_0\times I_{it}^{\vartheta} \tag{4}$$

$$\vartheta(I_{it})=\vartheta_0\times I_{it}^{\lambda} \tag{5}$$

where A_0,ϑ_0 represent benchmark efficiency; ϑ,λ represent the elasticities of information quality factors with respect to technical efficiency and environmental governance efficiency, respectively. $\vartheta>0,\lambda>0$; Substitute the above power function into the pollution function:

$$P_{it}=k\vartheta_0 Y_{it}^{\phi}E_{it}^{\psi}I_{it}^{-\lambda\eta} \tag{6}$$

Fifth, to analyze the elasticity of GTFP with respect to the information factor, we take the natural logarithm of the GTFP function and incorporate the power function of technological progress and the pollution function:

$$ln(GTFP_{it})=ln(A_0)+\vartheta ln(I_{it})-\delta(k\vartheta_0^{-\eta}Y_{it}^{\varphi}E_{it}^{\psi}I_{it}^{-\lambda\eta}) \tag{7}$$

Sixth, let $C=\delta k\vartheta_0^{-\eta}$ be a constant; Compute the partial derivative of a with respect to b to reveal the marginal effect of the information element:

$$\partial ln(GTFP_{it})/\partial I_{it}=\vartheta/I_{it}+C\,\lambda\,\eta Y_{it}^{\varphi}E_{it}^{\psi}I_{it}^{-\lambda\eta-1} \tag{8}$$

Due to $\partial ln(GTFP_{it})/\partial I_{it}>0$, the enhancement of information elements exerts a positive marginal effect on GTFP.

Additionally, based on the differentiation rules, we can obtain:

$$\partial ln(GTFP_{it})/\partial I_{it}=(1/GTFP_{it})\times(\partial GTFP_{it}/\partial I_{it}) \tag{9}$$

Seventh, this study introduces the impact of digital government construction DG_{it}. Digital government construction directly improves the quality of information elements. Suppose $I_{it}=I_0\times DG_{it}^{\omega}$ and take the derivative with respect to DG_{it}:

$$\partial I_{it}/\partial DG_{it}=I_0\times\omega DG_{it}^{w\text{-}1} \tag{10}$$

Due to $\partial I_{it}/\partial DG_{it}>0$, digital government construction directly enhances the quality of information elements.

Eighth, by combining Equations (8) and (9) with the chain rule, this study obtains:

$$\partial GTFP_{it}/\partial DG_{it}=(\partial GTFP_{it}/\partial I_{it})\times(\partial I_{it}/\partial DG_{it}) \tag{11}$$

Further organization yields:

$$\partial GTFP_{it}/\partial DG_{it}=GTFP_{it}\times(\vartheta/I_{it}+C\,\lambda\,\eta Y_{it}^{\varphi}E_{it}^{\psi}I_{it}^{-\lambda\eta-1})\times I_0\omega DG_{it}^{w\text{-}1} \tag{12}$$

Due to $\partial GTFP_{it}/\partial DG_{it}>0$, the digital government construction ultimately exerts a positive influence on GTFP by enhancing the quality of information elements. Therefore, this study proposes Hypothesis 1:

Hypothesis 1: The construction of digital government significantly promotes the improvement of GTFP.

3.2 Influence mechanism of digital government construction on GTFP

As the institutional carrier of digital government construction, the reform of big data management institutions first triggers an increase in the government's digital attention to green development. Establishing the government's big data bureau has forced the government to incorporate factors related to green total factor productivity into core decision-making indicators (such as visualizing carbon emission intensity, resource recycling rate and energy intensity through the data cockpit, etc.). It has prompted the policy agenda to tilt towards low-carbon goals (Xia, 2025). This institutional focus forms a "pressure-response" mechanism: higher-level governments strengthen accountability for green development through data performance evaluations, compelling local governments to tilt digital resources towards clean technology research and development, environmental compliance supervision, and other directions (Li et al., 2023). The institutionalized enhancement of digital attention has restructured the value ranking of urban development and laid a foundation for governance momentum for the improvement of green total factor productivity.

The deepening of government attention to digital figures further requires the support of new digital infrastructure to implement green governance. The Internet of Things perception network (such as ecological environment monitoring satellites and online monitoring of key pollution sources) and the city-level data middle platform, which the government's big data bureau comprehensively plans, have built a green digital foundation covering energy, transportation and industry (Zhou et al., 2022). Such facilities achieve real-time collection and intercommunication of environmental data throughout the life cycle, solving the defects of lagging ecological information and localization in the traditional mode (Janssen et al., 2009). For instance, smart grids dynamically match the supply and demand of clean energy, and intelligent transportation systems optimize vehicle traffic efficiency and reduce exhaust emissions. The digital infrastructure that is fully interconnected provides a physical foundation for the leap in green total factor productivity by reducing the cost of factor flow and information friction in the green transformation (Tang et al., 2025).

Supported by digital infrastructure, the data-sharing mechanism drives breakthroughs in core digital technologies. The government's open environmental databases (such as real-time emission data of enterprises and clean patent libraries) have reduced the uncertainty and trial-and-error costs of enterprises' digital technology research and development (Yu et al., 2025). The artificial intelligence analysis platform assists enterprises in optimizing the energy consumption of production processes and achieving the synergy of "data flow-technology flow" (Ahmad, 2025). For instance, enterprises rely on the intelligent manufacturing system of the industrial Internet to reduce the verification cycle of new materials through dynamic simulation models. Big data predictive maintenance reduces the energy consumption of enterprise equipment idling (Medaglia et al., 2024; Wang and Guo, 2024). The integrated innovation of digital and green technology has restructured the production function, replacing traditional factor stacking with total factor optimization (Ahmad, 2025). It has become the core application carrier driving GTFP.

The process eventually settles into the digital economy's systematic reshaping of the urban green development model. At the macro level, the market-based allocation of data elements (such as carbon data asset trading and green finance credit rating) guides capital to flow into low-carbon industries (Li and Yue, 2025). The platform economy reduces idle resources through the sharing model (such as smart logistics lowering the empty running rate), giving rise to new forms of green services. At the micro level, enterprises adjust their strategies based on the industry energy efficiency benchmarking data opened by the government, promoting the industrial chain to climb towards the high-value-added and low-carbon links of the "smile curve" (Guo et al., 2024). The transformation of the digital economy ecosystem brought about by digitalization, through optimizing the efficiency of factor combination and reconstructing the division of labor network in the value chain, systematically achieves the sustainable evolution of green total factor productivity improvement. Therefore, this study proposes Hypothesis 2:

Hypothesis 2: The digital government construction promotes the improvement of GTFP through chain influence mechanisms of governmental digital attention, digital infrastructure construction, digital technology innovation and digital economic development.

4 Research Design

4.1 Model Design

This study employs the partial linear double debiased machine learning difference-in-differences (DDML-DID) model specified in Equation (1), which effectively estimates the net treatment effect of policies while handling high-dimensional covariates and potential nonlinear relationships. By applying this robust causal inference approach, we accurately identify the impact of digital government construction on green total factor productivity (GTFP), providing refined and reliable empirical evidence for how digital governance enables green transition.

Furthermore, using residual-based instrumental variable regression, this study obtains an unbiased estimator of the policy treatment effect as formalized in Equation (3) (Chernozhukov et al., 2018). From a societal perspective, the rigorous methodological design allows policymakers to pinpoint key determinants—such as data integration capacity, regulatory enforcement efficiency, and institutional coordination—that drive successful green transformation. This evaluation framework offers actionable insights for public authorities to design more effective and sustainable digital governance strategies, ultimately supporting broader societal benefits including enhanced environmental quality, transparent policy-making, and inclusive low-carbon development.

$$GTFP_{it} = \theta_0 Policy_{it} + g(X_{it}) + U_{it} \tag{13}$$

$$E(U_{it} | Policy_{it}, X_{it}) = 0 \tag{14}$$

$$\hat{\theta}_0 = \left(\frac{1}{n}\sum_{i\in I,t\in T} \hat{V}_{it} Policy_{it}\right)^{-1} \frac{1}{n}\sum_{i\in I,t\in T} \hat{V}_{it}\left(GTFP_{it} - \hat{g}(X_{it})\right) \tag{15}$$

where $GTFP_{it}$ represents the dependent variable, which is the green total factor productivity; $Policy_{it}$ represents the independent variable, which is the big data management institution reform policy; $X_{it}$is a set of high-dimensional control variables and DDML algorithm is used to estimate its specific form $\hat{g}(X_{it})$; U_{it} represents the error term which conditional mean value is 0; θ_0 is the policy treatment coefficient that this study mainly focuses on; i represents city; *t* represents the year.

4.2 Variable measurement

4.2.1 Dependent variable

This study measures green total factor productivity (GTFP) using the Super-efficient SBM-Malmquist-Luenberger index method. Labor, capital, and energy are selected as input factors; urban economic development serves as the desirable output; and industrial sulfur dioxide, industrial smoke, and industrial wastewater are adopted as undesirable outputs. These three pollutants have long been recognized as key indicators of industrial pollution in China's environmental governance system and are closely linked to human health and ecological quality. Their inclusion aligns with the historical regulatory emphasis on conventional industrial pollutants and reflects policy priorities over the sample period (2007–2023). The calculation method of GTFP is as follows：

First, determine the directional distance function to handle the undesirable output measured by GTFP:

$$\vec{D}_t(x^t, y^t, b^t, g_y - g_b) = sup\{\beta: (y^t + \beta g_y, b^t - \beta g_b) \in P^t(X^t)\} \tag{16}$$

Second, the static efficiency value is calculated through the Super-efficient SBM model. This model allows the efficiency value to be greater than 1, which can handle the weak disposability of undesirable outputs and optimize the slack variables. Its directional function is as follows:

$$\rho^* = min\frac{1-\frac{1}{m}\sum_{i=1}^{m}\frac{s_i^-}{x_{io}}}{1+\frac{1}{s}(\sum_{r=1}^{s}\frac{s_r^+}{y_{ro}}+\sum_{K=1}^{P}\frac{s_k^b}{b_{ko}}} \tag{17}$$

Finally, the dynamic change rate is decomposed by the GML index. This index is based on the global production frontier and avoids the problem of infeasible solutions across periods. Its function is as follows:

$$GTFP_t^{t+1} = \sqrt{\frac{1+D_o^t(x^{t+1},y^{t+1},b^{t+1};y^{t+1},-b^{t+1})}{1+D_o^t(x^t,y^t,b^t;y^t,-b^t)} \times \frac{1+D_o^{t+1}(x^{t+1},y^{t+1},b^{t+1};y^{t+1},-b^{t+1})}{1+D_o^{t+1}(x^t,y^t,b^t;y^t,-b^t)}} \tag{18}$$

The distribution maps of GTFP in 2007 and 2023 are shown in Fig. 3. Overall, GTFP shows a global upward trend, and the coverage of the high-value area in Figure (b) in 2023 has significantly expanded. Although low-value agglomeration phenomena exist in regions such as Northeast and Northwest China, the GTFP advantages of core growth poles such as the Yangtze River Delta and the Pearl River Delta have been further strengthened, forming continuous high-value uplift belts. Regarding spatial dynamics, the growth of GTFP shows a diffusion pattern where "the coastal areas accelerate faster than the inland areas and the core urban agglomerations lead the surrounding areas", reflecting the coordinated advancement of the improvement of green total factor productivity and regional coordinated development. These distribution evolution trends provide an important factual basis for exploring the role of policy drivers such as digital government construction.

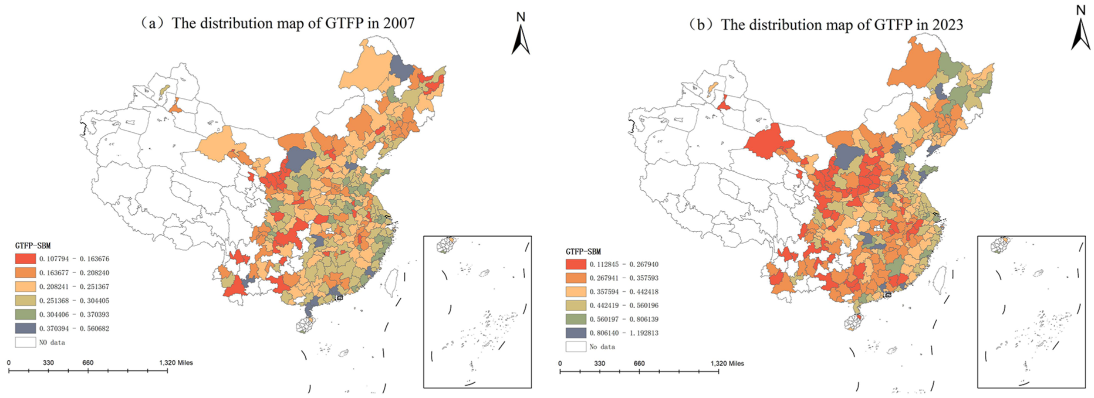

Fig. 3 The distribution maps of *GTFP* in 2007 and 2023

Notes: The maps are provided by the Ministry of Natural Resources of the People's Republic of China. The review number of maps is GS (2024) 0650.

4.2.2 Independent variable

This study measures the treatment effect of the big data management institution reform policy by the interaction term of the urban disposal type dummy variable and the policy implementation time (*Bdmi-policy*). This study takes the cities within the administrative divisions of 21 provinces that underwent the reform of big data management institutions in 2018, 2019 and 2021 as the experimental group, totaling 207 cities. The dummy variable of their urban disposal types is assigned as 1; The cities within the administrative divisions of the remaining provinces were taken as the control group, totaling 71 cities, and their urban disposal types are set to 0. After the reform of the big data management institution, the time dummy variable is assigned as 1, and vice versa, it is 0.

4.2.3 Control variable

Double debiased machine learning can effectively deal with high-dimensional control variables with its regularization algorithm. Therefore, to ensure the accuracy of policy effect estimation, based on existing studies and taking into account the availability of data, the following six variables are selected as possible control variables in this study referring to Zhou et al. (2025): Economic development (*LnGDP*); Financial development (*Finance*); Government intervention (*GOV*); Industrial structure (*Indus*); Opening-up (*Open*); Human capital (*HC*). In addition, quadratic terms of each city control variable are added to the regression analysis to improve the accuracy of the fitting model. Meanwhile, to avoid the loss of information on the dimensions of cities and time, the fixed effects of cities and time are introduced as individual and year dummy variables.

4.2.4 Mechanism variable

This study intends to reveal the mechanism effects of digital government construction in promoting regional green total factor productivity improvement from four paths: government digital attention, digital technology innovation, digital infrastructure construction and digital economic development.

Governmental digital attention (*DigAttention*) is quantified through text mining technology in the text information of the government work report. It is measured by the proportion of the keyword frequency related to the governmental digital attention (including two major aspects: digital technology and digital application, totaling 121 words) to the total keyword frequency. The greater the proportion of government digital word frequency in the total word frequency, the more government decision-makers pay attention to digital development issues and the higher the governmental digital attention. Similarly, digital infrastructure construction (*DigInfrus*) is measured by the proportion of the word frequency related to digital infrastructure in the government work report to the total word frequency of the text.

Digital technology innovation (*DigTech*) is measured by the number of invention patents related to digital technology applied by enterprises within the urban jurisdiction in the current year. Specifically, based on the "Reference Relationship Table between the Classification of

Strategic Emerging Industries and the International Patent Classification (2021)" released by the National Intellectual Property Administration, this study identifies the patents corresponding to IPC in digital industries such as big data, blockchain, and artificial intelligence as digital patents. It adds up the number of digital patents by year and city. Then, a logarithmic transformation will be performed to obtain digital technological innovation variables to alleviate the influence of patent data's skewed distribution problem on the policy effects' estimation coefficient.

Digital economic development (*DigEco*) is achieved by constructing a comprehensive development indicator system for the digital economy (including five aspects: Internet penetration rate, the number of Internet-related practitioners, Internet-related output, the number of mobile Internet users, and the inclusive development of digital finance), and using the principal component analysis method to standardize the data of the indicator system and then reduce its dimension.

4.2.5 Data sources and descriptive statistics

This study takes 278 prefecture-level cities and municipalities directly under the Central Government in China as the research sample, and the research period is from 2007 to 2023. The data in this study are sourced from the "China Urban Statistical Yearbook", the National Intellectual Property Administration, government work reports of various cities, the China Research Data Service Platform, etc. The missing values are filled in using the linear interpolation method, and cities with administrative division adjustments or severe data omissions have been excluded. The descriptive statistical results of the main variables are shown in Table 1.

Fig. 4's kernel density estimation results clearly illustrate the dynamic distribution evolution of GTFP across different groups under policy intervention. The full sample exhibits a multi-peak extended distribution pattern, indicating that overall GTFP growth exhibits phased transition characteristics. The kernel density curve for the treatment group shifts to the right with a significantly heightened peak, demonstrating that after policy implementation, GTFP not only increased overall but also showed a trend toward concentrated and strengthened growth momentum, confirming the positive promotional effect of digital governance on green productivity. The control group maintains a relatively stable distribution pattern with no noticeable rightward shift in the peak, indicating that groups unaffected by the policy continue along their original development trajectory.

Table 1 Descriptive statistics of the main variables

Variables	Obs.	Mean	Median	SD	Min	Max
GTFP-SBM	4726	0.335	0.307	0.139	0.103	1.193
GTFP-CCR	4726	0.570	0.544	0.172	0.219	2.510
Bdmi-policy	4726	0.236	0.000	0.425	0.000	1.000
LnGDP	4726	16.51	16.44	1.008	13.34	19.97
Indus	4726	46.04	46.33	11.25	11.59	90.97
GOV	4726	0.188	0.163	0.097	0.044	1.027
Finance	4726	1.018	0.838	0.644	0.075	12.82
Open	4726	0.190	0.081	0.314	0.000	3.384
HC	4726	0.019	0.010	0.024	0.000	0.147
DigEco	4726	0.053	0.038	0.075	0.000	0.94
DigInfrus	4726	0.154	0.126	0.135	0.000	1.381
DigTech	4726	0.532	0.541	0.228	0.000	1.216
DigAttention	4726	0.169	0.136	0.160	0.000	1.663

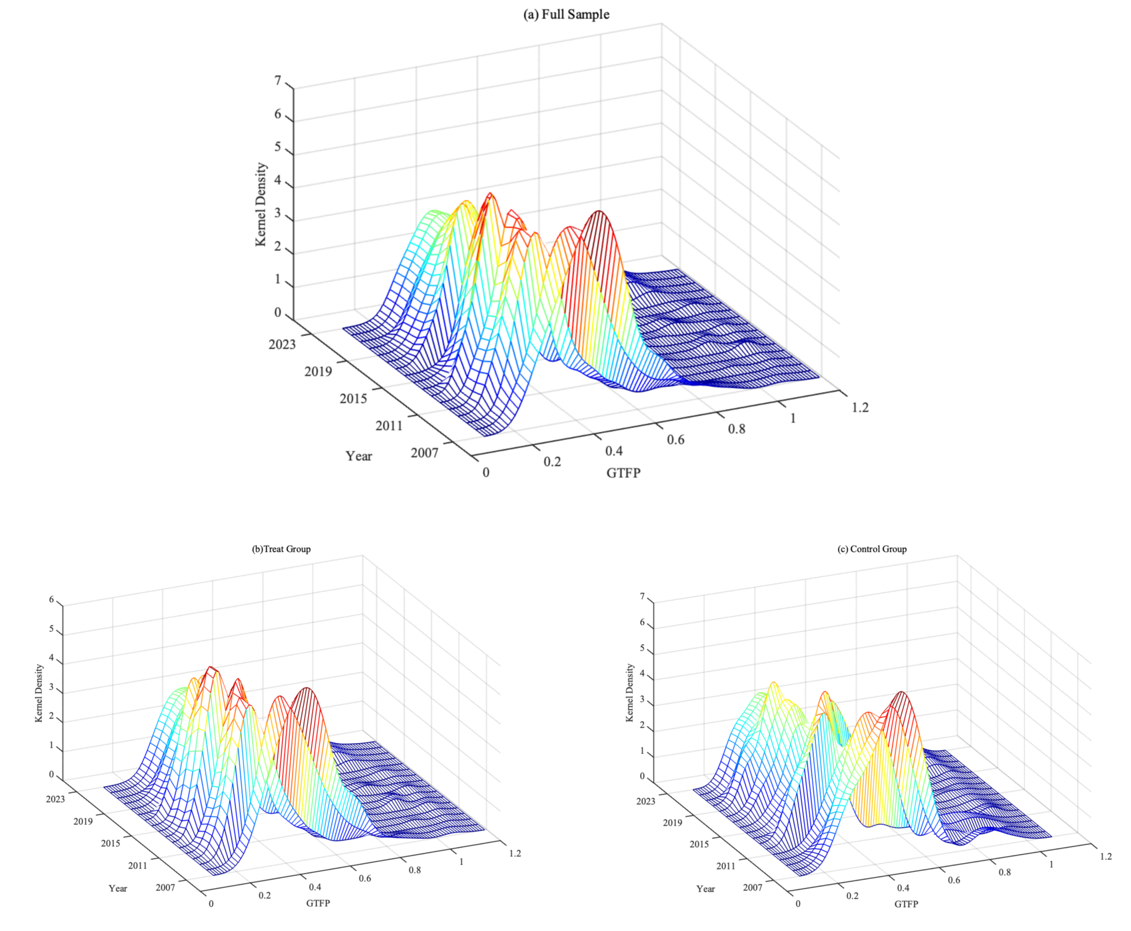

Fig. 4 The kernel density distributions of GTFP's values in various groups

5 Empirical results and analysis

5.1 The impact of digital government construction on GTFP

This study employs a double debiased machine learning approach to identify the causal effect of big data management institution reform—a core initiative in digital government construction—on urban green total factor productivity (GTFP). Multiple machine learning algorithms, including gradient boosting (GB), random forest (RF), neural network (NNT), support vector machine (SVM), and linear support vector machine (LSVM), were applied within a DDML-DID framework, all consistently producing positive and statistically significant estimates (at least at the 5% level) for the policy coefficient.

Specifically, the gradient boosting algorithm—selected for its superior performance in convergence and bias control (Yang et al., 2020)—yielded a coefficient of 0.019. This signifies that the implementation of digital government reform led to an average increase of approximately 1.9% in GTFP among treated cities. From a policy perspective, this result underscores that institutional innovation in data governance serves as a potent enabler of sustainable urban development. The empirical evidence suggests that breaking down administrative data silos through dedicated big data agencies can tangibly enhance green productivity, offering a replicable governance tool for policymakers aiming to balance economic growth with environmental targets.

Table 2 Benchmark regression results based on the partial liner model

Variables	*Various algorithms of double debiased machine learning*				
	GB	RF	NNT	SVM	LSVM
Bdmi-policy	0.019***	0.009**	0.047***	0.040***	0.034***
	(0.006)	(0.004)	(0.007)	(0.005)	(0.010)
Linear terms	YES	YES	YES	YES	YES
Quadratic terms	YES	YES	YES	YES	YES
Year Effects	YES	YES	YES	YES	YES
City Effects	YES	YES	YES	YES	YES
Observation	4726	4726	4726	4726	4726

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively; The values in parentheses are robust standard errors; Linear terms and Quadratic terms represent the linear terms and quadratic terms of the control variables, respectively.

5.2 Robustness test

5.2.1 Reset the DDML model

In this study, the benchmark regression is based on DDML to construct a partial linear model for analysis. The model setting is subjective. Thus, this study constructs a more general interactive model to alleviate the influence of model-setting bias on the research conclusion.:

$$GTFP_{it} = g(Policy_{it}, X_{it}) + U_{it} \tag{19}$$

$$Policy_{it} = m(X_{it}) + V_{it} \tag{20}$$

$$\hat{\theta} = E[g(Policy_{it} = 1, X_{it}) - g(Policy_{it} = 0, X_{it}) \tag{21}$$

In addition, due to data availability, this study inevitably has endogeneity problems such as omitted variables. Therefore, this study constructs a partial linear instrumental variable model of DDML and adopts the interaction term between the urban terrain undulation and the time trend as the instrumental variable. The specific settings are as follows:

$$GTFP_{it} = \theta_0 Policy_{it} + g(X_{it}) + U_{it} \tag{22}$$

$$Instrument_{it} = m(X_{it}) + V_{it} \tag{23}$$

where $Instrument_{it}$ represents the instrumental variable of $Policy_{it}$.

The rationality of choosing the terrain undulation and time trend term as instrumental variables is that the urban terrain undulation belongs to the natural geographical feature and is unrelated to any human factors, satisfying strict exogeneity. Second, the flatter the terrain of a city is, the more conducive it is to the construction and reform of big data management institutions, meeting the requirements of relevance. Third, since the terrain undulation belongs to the cross-sectional data, the time trend term (year-2006) is added to construct the panel data. The regression results of resetting the DDML model are shown in columns (1) and (2) of Table 3. After dealing with possible model missetting and endogeneity issues, the digital government construction still significantly promotes green total factor productivity, indicating that the conclusion of the benchmark regression in this study is robust.

5.2.2 Eliminate parallel policies' influence

In order to accurately assess the actual effectiveness of the big data management institution reform policy, it is necessary to eliminate the potential mutual influence of other parallel policies. Parallel policies related to this policy include the "National Big Data Comprehensive Pilot Zone" (*Bdce-policy*) and the "Smart City" pilot policy (*Scity-policy*), which aim to utilize digital technologies such as big data and artificial intelligence to promote the improvement of urban management efficiency, equalization of public services, and urban economic development. Therefore, in this study, the dummy variables and interaction terms (*Bdce-Scity*) of these two parallel policies are added to the DDML model for prediction and solution. The regression results are shown in columns (3) and (4) of Table 3. The results show that after controlling these two policies and their interaction terms respectively, the regression coefficient of the big data management institution reform policy on the green total factor productivity remains positively significant, proving the robustness of the research conclusion of this study.

5.2.3 Alter measure method of the independent variable

This study adopts the super-efficient SBM model to measure the green total factor productivity (*GTFP-SBM*). To further verify the robustness of the index measurement and research conclusions, the super-efficient CCR model is introduced to re-measure the green total factor productivity (*GTFP-CCR*) to eliminate the interference of excessive correction of slack variables. The regression results are shown in column (1) of Table 4. After changing the measurement method of the dependent variable, the digital government construction still has a significant positive effect on green total factor productivity, proving that the index measurement and research conclusion of this study are robust.

5.2.4 Add higher-dimensional fixed effects

Since provinces are critical administrative nodes in the governance structure of the Chinese government, cities under the same province often have similarities in terms of policy environment, resource endowment, historical culture, etc. Therefore, based on the partial linear model of DDML, this study adds the province-time interaction fixed effect to control the influence of changes in different provinces over time. The specific regression results are in column (2) of Table 4. After considering the correlation influence among the characteristics of different cities in the same province, the promoting effect of digital government construction on the growth of urban green total factor productivity is still significantly positive at the 1% level, and the conclusion of this study still holds.

5.2.5 Change model parameters

In this study, the model parameters are further adjusted to improve the utilization efficiency of data. The sample segments are set to 3 and 7 respectively, and the estimation is re-conducted.

The estimation results are shown in columns (3) and (4) of Table 4. The results show that after changing the sample segmentation ratio, the estimation coefficients of digital government construction for *GTFP* are all positively significant at the 1% level, indicating that the big data management institution reform policy has a significant promoting effect on the green total factor productivity of cities, proving the robustness of the benchmark regression results.

Table 3 Empirical results of the robustness test 1

Variables	*Reset DDML model*		*Eliminate parallel policies' influence*	
	Interactive model	*Instrumental variable*	*Single policy*	*Policy interaction*
Bdmi-policy	0.059**	0.223***	0.019***	0.020***
	(0.002)	(0.037)	(0.006)	(0.006)
Bdce-policy			YES	YES
Scity-policy			YES	YES
Bdce-Scity			NO	YES
Linear terms	YES	YES	YES	YES
Quadratic terms	YES	YES	YES	YES
Year Effects	YES	YES	YES	YES
City Effects	YES	YES	YES	YES
Observation	4726	4726	4726	4726

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively; The values in parentheses are robust standard errors; Linear terms and Quadratic terms represent the linear terms and quadratic terms of the control variables, respectively. The algorithm of DDML is Gradient Boosting.

Table 4 Empirical results of the robustness test 2

Variables	*Alter GTFP measure method*	*Add province -year effects*	*Change model parameters*	
			Kfolds=3	*Kfolds=7*
Bdmi-policy	0.014**	0.038***	0.020***	0.019***
	(0.007)	(0.007)	(0.006)	(0.006)
Linear terms	YES	YES	YES	YES
Quadratic terms	YES	YES	YES	YES
Year Effects	YES	YES	YES	YES
City Effects	YES	YES	YES	YES
Pro-year Effects	NO	YES	NO	NO
Observation	4726	4726	4726	4726

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively; The values in parentheses are robust standard errors; Linear terms and Quadratic terms represent the linear terms and quadratic terms of the control variables, respectively. The algorithm of DDML is

Gradient Boosting.

5.3 Mechanism analysis

5.3.1 Traditional mechanism analysis

According to the previous theoretical analysis, this study selects governmental digital attention (*DigAttention*), digital technology innovation (*DigTech*), digital infrastructure construction (*DigInfus*) and digital economic development (*DigEco*) as possible mechanism variables for digital government construction to boost GTFP. The mechanism test results are shown in Table 5.

The regression results in Table 5 show that the estimated coefficients of *Bdmi-policy* on *DigAttention, DigTech, DigInfrus* and *DigEco* are all significantly positive, indicating that there are four significant mechanism paths for the promotion effect of digital government construction on GTFP, which is consistent with Hypotheses 2. These mechanisms do not operate in isolation but form a sequential and reinforcing chain. Specifically, governmental digital attention serves as the initial trigger, creating administrative momentum for data-driven governance. This leads to investment in digital infrastructure, which provides the physical backbone for data integration and smart management. These infrastructures, in turn, facilitate digital innovation—notably in green technologies—as improved data access lowers R&D costs and uncertainty. Ultimately, these technical and institutional advancements coalesce into broader digital economic development, optimizing resource allocation and fostering new green business models.

Notably, *DigAttention* is the most responsive indicator to *Bdmi-policy,* indicating that in the initial stage of policy implementation, the improvement of GTFP was mainly driven by administrative forces, such as through assessment and accountability, and the establishment of data openness mechanisms, which quickly generated governance momentum. In contrast, *DigTech* shows a significant lag effect. Its regression coefficient is relatively small (0.012), reflecting that technology needs to go through processes such as learning and digestion, application adaptation, and market diffusion before generating productivity gains. Therefore, the contribution of GTFP to digital technological innovation will take a longer time to fully manifest.

The heterogeneity of response times in this mechanism provides important insights for policy implementation. In the early stage of the policy, efforts can be prioritized by enhancing the government's digital attention to quickly establish institutional foundations and data governance systems, achieving visible results in the short term. However, in the medium and long term, attention should be paid to cultivating technological innovation and transformation capabilities, especially focusing on the differences in technology absorption and application among different cities. For instance, non-resource-based cities and central cities have better innovation foundations and shorter technology diffusion cycles; while resource-based cities and peripheral cities are constrained by innovation resources and infrastructure, and the effect of technology takes significantly longer to manifest. Therefore, the policy implementation should design a dynamic temporal sequence: in the early stage, administrative promotion and facility layout should be emphasized; in the middle and later stages, support for technological research and development should be increased and differentiated promotion strategies should be implemented to adapt to the response rhythms and capacity characteristics of different cities, ensuring the sustainability and inclusiveness of the policy effects.

Table 5 Empirical results of the mechanism tests

Variables	*GTFP-SBM*	*DigAttention*	*DigTech*	*DigInfrus*	*DigEco*
Bdmi-policy	0.019***	0.032***	0.012***	0.015**	0.019***
	(0.006)	(0.008)	(0.004)	(0.006)	(0.006)
Linear terms	YES	YES	YES	YES	YES
Quadratic terms	YES	YES	YES	YES	YES
Year Effects	YES	YES	YES	YES	YES
City Effects	YES	YES	YES	YES	YES
Observation	4726	4726	4726	4726	4726

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively; The values in parentheses are robust standard errors; Linear terms and Quadratic terms represent the linear terms and quadratic terms of the control variables, respectively. The algorithm of DDML is Gradient Boosting.

5.3.2 Causal mechanism analysis

To further examine the robustness and heterogeneity in the mechanisms through which digital government construction affects GTFP, this study employs causal mechanism analysis within a counterfactual framework. This approach disentangles the intertwined relationships between treatment status, mediating pathways, and outcomes by separating the direct effect of the policy intervention from its indirect effects operating through each mechanism. This allows for a more nuanced understanding of how initial conditions and contextual factors shape the efficacy of each pathway.

The decomposition results in Table 6 reveal that while the total and direct effects of digital government construction on GTFP are significantly positive across the sample, notable heterogeneity emerges in the indirect effects across treatment and control groups, highlighting the context-dependent nature of each mechanism. Governmental Digital Attention (*DigAttention*): A significant positive indirect effect (0.018) is observed in the control group, whereas a negative but insignificant effect (-0.009) appears in the treat group. These phenomenon suggest that in non-pilot cities, increased digital attention—often driven by policy spillover and emulation—effectively promotes GTFP by breaking initial information barriers and activating administrative momentum. However, in pilot cities where digital governance systems are more mature, further increases in digital attention may yield diminishing returns. Institutional saturation may set in, where additional emphasis on data-driven governance crowds out other productive inputs or leads to redundancy in regulatory efforts, thereby attenuating its marginal contribution to GTFP.

Digital Infrastructure Construction (*DigInfus)*: The significant indirect effect in the control group (0.024) underscores that cities with inadequate infrastructure benefit markedly from new investments in data platforms and IoT networks, which reduce information asymmetry and strengthen environmental oversight. In contrast, the treat group shows an insignificant effect (0.005), implying that cities that already possess advanced digital infrastructure face diminishing returns. Further investments may encounter physical or technical ceilings, such as limited scalability of existing systems or interoperability challenges, requiring breakthroughs in next-generation technologies to unlock new GTFP gains.

Digital Technology Innovation (*DigTech*): A strong and significant indirect effect is detected in the treat group (0.026), while the effect is negligible in the control group (0.005). This divergence highlights the critical role of pre-existing innovation ecosystems. Pilot cities, equipped with policy advantages such as data access permissions and R&D subsidies, possess stronger absorptive capacity. They can more rapidly convert digital technologies into productive applications, enabling tangible gains in green productivity. Non-pilot cities, however, often lack the complementary assets — such as skilled human capital, industry-academia linkages, or risk capital—necessary to assimilate and deploy advanced technologies, resulting in weak translational momentum.

Digital Economic Development (*DigEco*): The indirect effects are statistically insignificant in both groups, though slightly larger in pilot cities. This indicates that the transition to a digital economy is a complex, systemic process that cannot be driven by government action alone. It requires synchronized reforms in market mechanisms, factor allocation, and private-sector participation. The modest effect sizes suggest that while digital government construction initiates the transition, its full fruition depends on deeper coordination between state-led digitalization and market-led entrepreneurial activity.

These heterogeneous mechanisms emphasize that the impact of digital government construction is not uniform but contingent on local conditions and pre-existing institutional and technological capabilities. Effective policy design should therefore be tailored and sequential: leveraging quick wins through attention and infrastructure in less-developed cities, while focusing on technology adoption and ecosystem development in more advanced regions.

Table 6 Empirical results of the mechanism decomposition

Variable	Total effect	*Control group*		*Treat group*	
		Direct effect	Indirect effect	Direct effect	Indirect effect
DigAttention	0.069***	0.078***	0.018**	0.051***	-0.009
	(0.015)	(0.014)	(0.007)	(0.016)	(0.015)
DigInfrus	0.069***	0.064***	0.024***	0.045***	0.005
	(0.015)	(0.011)	(0.006)	(0.005)	(0.013)
DigTech	0.069***	0.043***	0.005	0.064***	0.026*
	(0.015)	(0.014)	(0.011)	(0.017)	(0.066)
DigEco	0.069***	0.065***	0.002	0.067***	0.030
	(0.015)	(0.014)	(0.009)	(0.017)	(0.012)

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively; The values in parentheses are robust standard errors; The algorithm of DDML is Gradient Boosting. The estimation results are obtained by programming based on the RStudio's Causalweight package.

5.4 Heterogeneity analysis

5.4.1 Ecological function

Ecological function zones (EFZs) are regions designated to serve key ecological functions—such as soil and water conservation and biodiversity protection—where restrictions apply on large-scale industrial and urban expansion to sustain ecological product supply. To examine whether the big data management reform policy exerts heterogeneous effects across cities with varying ecological responsibilities, this study refers to China's National Main Function Zone Planning and classifies cities into ecological function zones (EFZ) and non-ecological function zones (NEFZ) for subgroup regression. Results are presented in columns (1) and (2) of Table 7.

Empirical results show that the promoting effect of the big data management institution reform policy on GTFP is more significant in ecological function zones, but not in non-ecological function zones. This divergence can be attributed to fundamental differences in institutional constraints and policy urgency. EFZs operate under stringent ecological redlines and face binding environmental targets, which amplify the value of digital governance tools. Technologies such as real-time pollution monitoring, cross-departmental data sharing, and intelligent environmental compliance systems allow these regions to overcome information asymmetry and enforce regulations with higher precision—directly alleviating historical bottlenecks in ecological resource allocation.

In contrast, NEFZs experience weaker environmental regulatory pressure and possess greater industrial flexibility. Without strong ecological performance mandates, the marginal benefit of digital governance remains limited. Moreover, the economic structure in many NEFZs is less resource- or ecologically-bound, reducing the perceived need for data-intensive environmental oversight. Consequently, the introduction of digital government measures fails to trigger substantial green productivity gains in these areas.

5.4.2 Resource endowment

Cities with resource advantages often develop a mono-industrial structure, leading to long-term lock-in effects and path dependence on high-investment, extensive-processing industrial models. Moreover, resource-intensive industries frequently divert excessive capital and labor away from greener sectors, inhibiting green total factor productivity (GTFP) and creating a "resource curse". To examine how digital government construction differentially affects GTFP across resource contexts, this study classifies sample cities into resource-based cities (RBCs) and non-resource-based cities (NRBCs) according to the National Sustainable Development Plan for Resource-Based Cities (2013–2020), and conducts subgroup regression. Results are presented in columns (3) and (4) of Table 7.

The regression results indicate a pronounced heterogeneity in policy effects based on urban resource endowment: resource-based cities (RBCs) demonstrate significantly stronger responsiveness to the big data management institution reform, whereas non-resource-based cities (NRBCs) exhibit comparatively weaker policy sensitivity.This divergence can be attributed to structural and institutional factors specific to RBCs. Traditionally constrained by the "resource curse," these cities often exhibit locked-in development patterns characterized by energy-intensive industries and high pollution path dependence. Digital government construction helps break this path dependency by introducing data-driven governance tools—such as energy consumption monitoring platforms, cross-sector environmental data sharing, and intelligent compliance systems—which significantly enhance the precision and enforcement of environmental regulations. By mitigating information asymmetry and optimizing resource allocation, digital governance empowers RBCs to decouple economic growth from pollution emissions and accelerate green upgrading of traditional industries.

In contrast, NRBCs generally possess more diversified and lighter industrial structures, with lower energy intensity and fewer legacy pollution issues. The already relatively advanced production structure offers limited potential for further green productivity gains through digital administrative interventions alone. Consequently, the marginal effect of digital government construction appears weaker in these regions.

5.4.3 Environmental target

The two control zones refer to the acid rain and sulfur dioxide control zones. As strict environmental regulation zones in China, the two control zones shoulder significant responsibilities for transforming high-carbon industries, achieving dual carbon goals, and sustainable development. To explore the possible differentiated impacts of urban environmental targets on digital government construction, this study divides the city into two-control zones (TCZ) and non-two-control zones (NTCZ) for regression based on the

"Acid Rain Control Zone and Sulfur Dioxide Pollution Control Zone Division Scheme" proposed by the State Environmental Protection Administration of China. The regression results are shown in columns (5) and (6) of Table 7.

The empirical results reveal a counterintuitive yet significant heterogeneity in the impact of digital government construction on GTFP across cities with different environmental regulation intensities. Specifically, the policy effect is significantly positive in non-two-control zones (NTCZs), whereas it is negative (though statistically insignificant) in two-control zones (TCZs)—a finding that appears to contradict theoretical expectations.

This paradox can be explained through institutional interplay and policy crowding-out effects. TCZs, which are subject to stringent pre-existing environmental mandates—such as total emissions caps and mandatory desulfurization targets—already operate under a dense regulatory framework. The introduction of digital government initiatives, while conceptually complementary, may in practice lead to implementation overlap and regulatory redundancy. For instance, newly established data monitoring systems could duplicate existing compliance review processes, increasing administrative transaction costs without substantially enhancing environmental oversight. Moreover, the cognitive and bureaucratic load associated with adopting parallel digital and conventional systems might dilute enforcement effectiveness and weaken coordination across agencies, ultimately attenuating the net policy benefit.

Conversely, NTCZs typically lack such intensive command-and-control regulations, resulting in a relative environmental governance gap. Here, digital government construction acts as a foundational institutional upgrade: it introduces data-driven oversight mechanisms where they were previously absent, strengthens environmental compliance through intelligent monitoring and cross-department data integration, and significantly reduces information asymmetry in pollution management. Thus, the marginal effect of digital governance is markedly higher, generating a clear and positive impact on GTFP.

5.4.4. Urban grade

Since municipalities directly under the Central Government, provincial capitals and sub-provincial cities are the centers of regional economy, politics and culture, the digital government construction may have differentiated characteristics compared with other cities, leading to different impacts on green total factor productivity. Therefore, in this study, municipalities directly under the Central Government, provincial capitals and sub-provincial cities are classified as Central cities (Center), and the others are classified as Marginal cities (Margin), and group regression analysis is conducted. The regression results are shown in columns (7) and (8) of Table 7.

The regression results indicate a pronounced heterogeneity in the treatment effect of the big data management institution reform policy across urban hierarchies: central cities exhibit a significantly higher policy response relative to marginal cities. This divergence stems from profound disparities in institutional capacity, resource endowments, and technological absorption capabilities. Central cities, benefit from concentrated administrative resources, including advanced digital infrastructure (e.g., provincial data hubs and cloud computing platforms), denser clusters of high-skilled talent, and more mature innovation ecosystems. These advantages enable efficient policy implementation and rapid technology diffusion, allowing digital governance tools to be seamlessly integrated into existing environmental regulatory frameworks. The agglomeration of political and economic resources further amplifies coordination across departments, enhances data-sharing efficiency, and accelerates the translation of policy directives into tangible green productivity gains.

In contrast, marginal cities often operate with fragmented digital infrastructure, limited fiscal capacity, and thinner human capital reserves. Constraints in broadband penetration, data storage facilities, and technical expertise hinder the effective deployment and operation of big data systems. Moreover, weaker institutional networks and lower organizational adaptability reduce the speed and fidelity of policy transmission, resulting in diminished marginal returns on digital governance investments.

Table 7 Empirical results of the heterogeneity tests

Variable	EFZ	NEFZ	RBC	NRBC	TCZ	NTCZ	Center	Margin
Bdmi-policy	0.018^{***}	0.005	0.028^{***}	0.014^{*}	-0.001	0.023^{***}	0.045^{**}	0.010^{*}
	(0.007)	(0.009)	(0.008)	(0.008)	(0.007)	(0.008)	(0.019)	(0.005)
Linear terms	YES	YES	YES	YES	YES	YES	YES	YES
Quadratic terms	YES	YES	YES	YES	YES	YES	YES	YES
Year Effects	YES	YES	YES	YES	YES	YES	YES	YES
City Effects	YES	YES	YES	YES	YES	YES	YES	YES
Observation	2210	2516	1938	2788	2669	2057	595	4131

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively; The values in parentheses are robust standard errors; Linear terms and Quadratic terms represent the linear terms and quadratic terms of the control variables, respectively. The algorithm of DDML is Gradient Boosting.

5.5 Further discussion: Individual treatment effects based on GRF

The generalized random forest (GRF) algorithm, as a non-parametric policy evaluation method, estimates the individual treatment effect by constructing an adaptive neighborhood weight, thereby effectively dealing with the estimation caused by excessive covariates, unbalanced distribution, and insufficient time-varying in traditional causal inference models such as propensity score matching and difference-in-differences models, and significantly improving the credibility of policy effect estimation (Athey et al., 2019).

Thus, this study estimates the individual policy treatment effect as shown in formula (16) of the big data management institution reform policy on GTFP based on the GRF algorithm. This study draws the effect distribution diagram as shown in Fig. 5. The distribution of individual treatment effects of GRF indicates that under different tree sizes (100 to 10000 trees), the conditional average treatment effect (CATE) of digital government construction on GTFP shows a right-skewed distribution. Its core mode interval is stably distributed within the positive interval of [0.00, 0.10] (taking 10,000 trees in (d) as an example, the frequency peak reaches over 1000), which rigorously confirms the core conclusion in the benchmark regression that "the construction of digital government has a significant promoting effect on green total factor productivity".

Notably, as the forest size expanded (from 100 trees to 10000 trees), the dispersion estimated by CATE decreased significantly (spanning -0.10 to 0.30 and converging to -0.10 to 0.20). The variance convergence process conforms to the asymptotic consistency theory described by Athey et al. (2019) - that is, the increase in the number of trees optimizes the local moment estimation accuracy through the gradient splitting rule, thereby alleviating the overfitting risk of small sample forests. However, a small amount of negative effect estimation occurred when the tree size was small (the left long tail of -0.05), when the number of trees increased to a sufficient size, the negative effect tail disappeared (only a trace distribution of -0.05 to 0.00 remained). This distribution pattern reveals continuous gradient differences in the policy effects of digital government construction at the micro-individual level. Still, its direction of action is always positive, which provides a non-parametric inference basis for accurately identifying policy-sensitive groups.

$$\hat{\tau}(\cdot) = argmin_{\tau}\{\textstyle\sum_{i=1}^{n}[Y_i - \widehat{m}^{(-i)}(X_i)] - \tau(X_i)[D_i - \hat{e}^{(-i)}]^2 + \varpi_n[\tau(\cdot)]\} \quad (24)$$

where $\hat{\tau}(\cdot)$ is the individual policy treatment coefficient under the condition of a given confusion variable X_i; $\widehat{m}^{(-i)}$ is the predicted value of the policy effect Y_i under the condition of a given confusion variable X_i out-of-Bag; $\hat{e}^{(-i)}$ is the probability of an individual accepting the treatment under the condition of a given confusion variable X_i out-of-Bag; D_i is the treatment variable; $\varpi_n[\tau(\cdot)]$ is the regularization term.

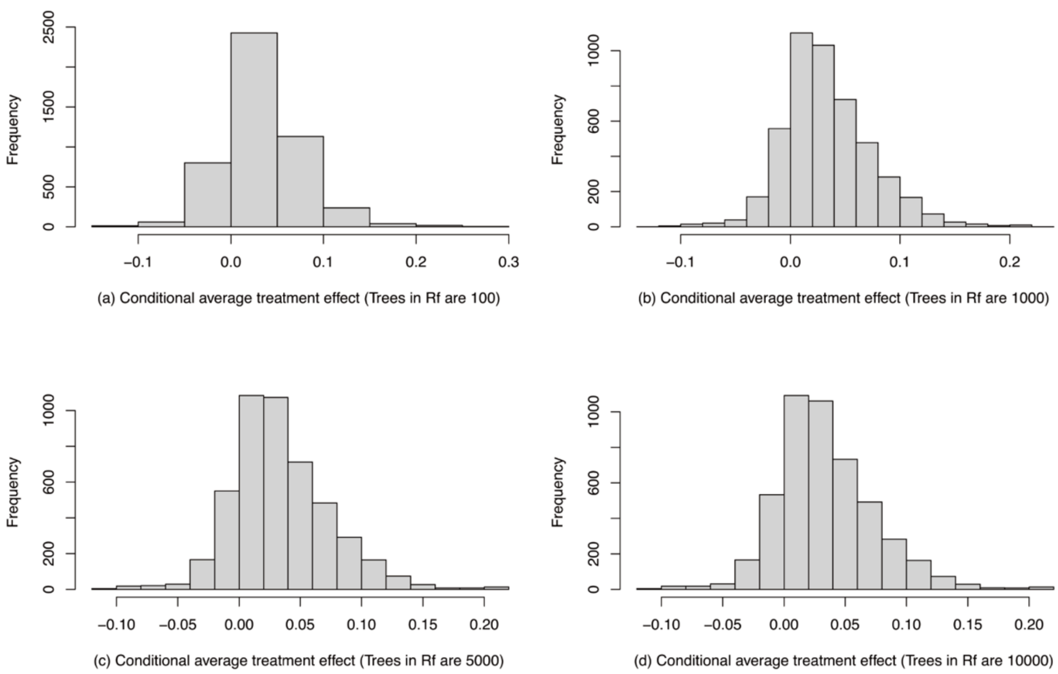

Fig. 5 The CATE distribution of digital government construction on GTFP

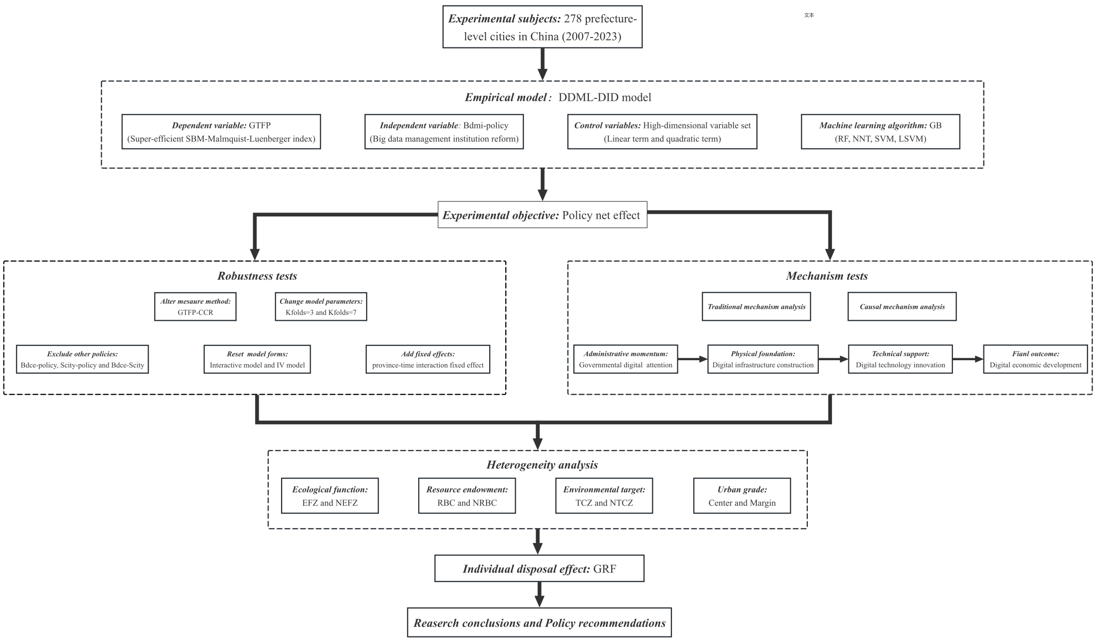

Fig. 6 Methodological framework

6 Research conclusions and policy recommendations

6.1 Research conclusions

The global digital revolution is fundamentally reshaping government governance capabilities, offering new pathways to digitally empower urban green transformation. Grounded in the "information-power" theory from political science, this study addresses the core research problem of how to overcome the entrenched "data island" phenomenon within traditional bureaucratic systems—which leads to fragmented environmental information, weakens regulatory enforcement, and ultimately constrains green total factor productivity (GTFP). To tackle this issue, this study conceptualizes the reform of big data management institutions as a quasi-natural experiment and employ a panel dataset of 278 Chinese prefecture-level cities from 2007 to 2023. Using a DDML-DID model, this study rigorously identifies the causal impact of digital government construction on GTFP.

The results demonstrate that digital government construction significantly promotes GTFP. This finding remains robust across multiple machine learning algorithms and an array of sensitivity checks. Mechanism analysis reveals that this promotion effect operates through four interlinked channels: First, by elevating governmental digital attention, which reorients policy agendas toward green goals; Second, by accelerating digital infrastructure development, which enables real-time environmental data integration and smart management; Third, by stimulating digital technological innovation, which enhances green production efficiency and reduces resource waste; Fourth, by fostering the growth of the digital economy, which optimizes resource allocation and fosters green business models. These mechanisms exhibit varying degrees of policy sensitivity between treatment and control groups, underscoring the context-dependent nature of digital governance effects. Furthermore, heterogeneity analysis indicates that the positive impact is more pronounced in ecological function zones, resource-based cities, non-two-control zones, and central cities, highlighting the importance of tailored policy implementation based on urban characteristics. Finally, using the GRF method, we uncover a continuous gradient distribution of positive individual treatment effects across most cities, reinforcing the generalizability and practical relevance of the big data management institution reform.

Theoretical contributions of this study are threefold: First, this study introduces political science's "information-power" theory into environmental economics, developing an interdisciplinary transmission logic that connects institutional innovation to productivity gains; Second, this study advances empirical methodology through the application of DDML-based causal inference in a staggered policy setting, mitigating biases from non-random adoption and complex variable relationships; Third, this study reveals a dynamic, multi-level mechanistic structure that enriches the literature on how digital governance transforms sustainable development outcomes. These insights not only deepen the understanding of the political economy of green transformation but also provide a validated analytical framework for future research in digital government and environmental governance.

6.2 Policy recommendations and research limitations

The research findings of this study are derived from China's unique institutional context—a hierarchical, top-down governance system with strong policy implementation capacity and significant government-led investment in digital infrastructure. Such a context allows for rapid scaling of reforms such as the big data management institutional reform, which effectively broke down administrative data silos and enhanced information-based environmental governance. These elements may serve as a valuable reference for other developing countries facing similar challenges, including fragmented environmental governance, insufficient regulatory enforcement, and delayed digital transformation.

First, implement differentiated data governance frameworks tailored to urban types. In ecological functional zones and resource-based cities—where digital government interventions proved most effective—develop mandatory environmental data sharing platforms and real-time pollution monitoring systems. Establish clear data ownership and interoperability standards to integrate dispersed environmental, energy, and industrial data. In less-developed marginal cities, focus on basic digital infrastructure and data literacy programs to prevent a “green digital divide”.

Second, introduce tiered performance metrics integrating digital governance with ecological targets. Leverage the finding that government digital attention significantly drives green productivity by embedding eco-specific indicators—such as carbon intensity, pollution monitoring accuracy, and data openness—into the evaluation systems of local officials. In central and resource-rich cities, emphasize innovation-led green outcomes; in ecological zones, prioritize regulatory enforcement and data-enabled oversight.

Third, foster innovation ecosystems through place-based digital policy. Where the study found strong effects of digital technology innovation—particularly in central cities and ecological zones—governments should establish open digital R&D platforms, share public patents, and offer sandbox mechanisms for green tech experimentation. In resource-based cities, support digital transformation of traditional industries via IoT and AI-driven upgrading, aligning with their industrial structure and path-dependent advantages.

Fourth, adopt context-sensitive financing instruments for digital-green transitions. Develop green digital finance mechanisms suited to local economic structures: in ecologically sensitive and resource-dependent regions, introduce data-backed green credit systems and carbon asset trading; in less-developed regions, facilitate targeted fiscal transfers and international green funds for scalable digital infrastructure.

Notably, this study has limitations that require further exploration. First, the measurement of the core dependent variable. In this study, CO2 is excluded as an undesirable output in GTFP measurement primarily due to conceptual and policy alignment. China’s environmental governance during the sample period prioritized conventional industrial pollutants for their direct local health and ecological impacts, while CO2 remained largely unregulated until the post-2020 “dual carbon” era. Methodologically, CO2 emissions exhibit high collinearity with energy input, risking double-counting and distorted efficiency estimation in production-theoretic models. Nonetheless, this omission constitutes a key limitation. As carbon neutrality gains strategic prominence, future research should integrate CO2 to better capture comprehensive environmental impacts and align GTFP assessment with evolving climate objectives.

Second, applicability of research findings to developed economies. Although the research findings and policy recommendations in this paper are applicable to some extent for developing countries with moderate governance capacity and centralized governance characteristics, their applicability to developed countries may be limited. Developed economies typically operate within multi-tiered governance frameworks, possessing mature market mechanisms, robust regulatory environments, and well-established digital ecosystems. The challenges they face often involve integrating traditional systems with new technologies within already decentralized environments and managing cross-departmental coordination—issues that are not central to this study. How to unleash the green production effects of digital government in developed economies warrants further exploration.

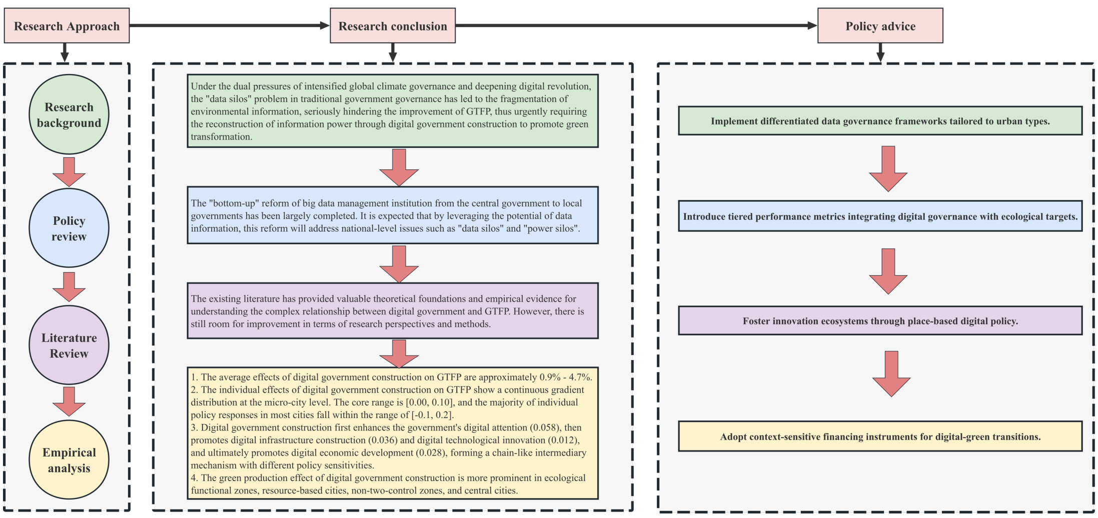

Fig. 7 Research conclusion framework

Funding This research is supported by the 17th Yunnan University Graduate Student Research Innovation Fund Project (KC-252513614).

Data availability Data will be made available on reasonable request.

Declarations

Consent to participate The authors consent to participate in the research.

Consent for publication The authors consent to publish the paper.

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Athey, S., Tibshirani, J., & Wager, S. (2019). Generalized random forest. *The Annals of Statistics, 47*(2), 1148-1178. https://doi.org/10.1214/18-AOS1709.

Ahmad, S. (2025). Pathways to environmental sustainability in developing Asia: Insights from environmental policy, green innovation and energy structure. *Renewable and Sustainable Energy Reviews, 216,* 115646. https://doi.org/10.1016/j.rser.2025.115646.

Ahmad, S. (2025). Green technologies, R&D and environmental policies: a dual ARDL-NARDL approach to sustainable transition. *Clean Technologies and Environmental Policy*, 1-22. https://doi.org/10.1007/s10098-025-03279-6.

Chen, L., Dai, T., Zhang, C., & Zhang, Z. (2025). Digital government and corporate ESG performance. *International Review of Financial Analysis, 104379*. https://doi.org/ 10.1016 /j.irfa.2025.104379.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal, 21*(1), C1-C68. https://doi.org/10.1111/ectj.12097.

Dahl, R. A. (2007). On political equality. Yale University Press.

Feng, Y., Han, Y., Hu, S., & Pan, Y. (2025). Alleviating energy poverty globally: Does digital government matter?. *Energy Economics, 143,* 108272. https://doi.org/10.1016/j.eneco. 2025.108272.

Gao, D., Li, G., & Yu, J. (2022). Does digitization improve green total factor energy efficiency? Evidence from Chinese 213 cities. *Energy, 247,* 123395. https://doi.org/10.1016 /j.energy.2022.123395.

Gong, Y., & Li, X. (2023). Designing boundary resources in digital government platforms for collaborative service innovation. *Government information quarterly, 40*(1), 101777. https:/ /doi.org/10.1016/j.giq.2022.101777.

Guo, P., Yin, Z., Zhu, H., & Gou, K. (2024). The Impact of Governments' Digital Economy Procurement on Rural Household Entrepreneurship. *Emerging Markets Finance and Trade, 60*(12), 2703-2718. https:/ /doi.org/10.1080/1540496X.2024.2308182.

Huang, M., Cheng, P., & Yuan, Y. (2025). Empowering corporate growth: How digital government enhances investment stability. *Economic Analysis and Policy*. https://doi.org/ 10.1016/j.eap.2025.06.004.

Jiakui, C., Abbas, J., Najam, H., Liu, J., & Abbas, J. (2023). Green technological innovation, green finance, and financial development and their role in green total factor productivity: Empirical insights from China. *Journal of Cleaner Production, 382,* 135131. https://doi.org/ 10.1016/j.jclepro.2022.135131.

Janowski, T. (2015). Digital government evolution: From transformation to contextualization. *Government information quarterly, 32*(3), 221-236. http://dx.doi.org/ 10.1016/j.giq.2015.07.001.

Jiang, Z., Yuan, C., & Xu, J. (2024). The impact of digital government on energy sustainability: Empirical evidence from prefecture-level cities in China. *Technological Forecasting and Social Change, 209,* 123776. https://doi.org/10.1016/j.techfore.2024. 123776.

Jiang, Y. (2025). Government digital governance and corporate green total factor productivity. *International Review of Economics & Finance, 104338*. https://doi.org/ 10.1016/j.iref.2025.104338.

Janssen, M., Chun, S. A., & Gil-Garcia, J. R. (2009). Building the next generation of digital government infrastructures. *Government Information Quarterly, 26*(2), 233-237. https://doi.org/ 10.1016/j.giq.2008.12.006.

Lyu, Y., Wang, W., Wu, Y., & Zhang, J. (2023). How does digital economy affect green total factor productivity? Evidence from China. *Science of the Total Environment, 857,* 159428. http://dx.doi.org/10.1016/j.scitotenv.2022.159428.

Lee, C. C., & Lee, C. C. (2022). How does green finance affect green total factor productivity? Evidence from China. *Energy economics, 107,* 105863. https://doi.org/10.1016/j.eneco. 2022.105863.

Li, B., & Wu, S. (2017). Effects of local and civil environmental regulation on green total factor productivity in China: A spatial Durbin econometric analysis. *Journal of Cleaner Production, 153,* 342-353. http://dx.doi.org/10.1016/j.jclepro.2016.10.042.

Lin, B., & Chen, Z. (2018). Does factor market distortion inhibit the green total factor productivity in China?. *Journal of Cleaner Production, 197,* 25-33. https://doi.org/10.1016 /j.jclepro.2018.06.094.

Li, C., Chen, X., & Yuan, C. (2025). Does digital government reduce carbon emissions? Empirical evidence from global sources. *Journal of Environmental Management, 380,* 125081. https://doi.org/10.1016/j.jenvman.2025.125081.

Li, E., Chen, Q., Zhang, X., & Zhang, C. (2023). Digital government development, local governments' attention distribution and enterprise total factor productivity: Evidence from China. *Sustainability, 15*(3), 2472. https://doi.org/10.3390/su15032472.

Li, X., & Yue, S. (2025). Does the government digital attention improve China's digital economy output efficiency: Accelerator or inhibitor. *Economic Analysis and Policy, 85,* 607-625. https://doi.org/10.1016/j.eap.2024.12.020.

Medaglia, R., Rukanova, B., & Zhang, Z. (2024). Digital government and the circular economy transition: An analytical framework and a research agenda. *Government Information Quarterly, 41*(1), 101904. https://doi.org/10.1016/j.giq.2023.101904.

Peeters, R., Miller, S. M., & Schuilenburg, M. (2025). Digital government inclusion: Exploring strategies for inclusive government automation. *Government Information Quarterly, 42*(2), 102028. http://dx.doi.org/10.1016/j.giq.2015.07.001.

Song, M., Du, J., & Tan, K. H. (2018). Impact of fiscal decentralization on green total factor productivity. *International Journal of Production Economics, 205,* 359-367. https://doi.org/ 10.1016/ j.ijpe.2018.09.019.

Tang, J., Li, W., Hu, J., & Ren, Y. (2025). Can government digital transformation improve corporate energy efficiency in resource-based cities?. *Energy Economics, 141,* 108043.

https://doi.org/10.1016/j.eneco.2024.108043.

Tan, W., Zhang, Z., Chu, Z., & Chen, P. (2024). Unlocking green potential: The digital government-driven revolution in corporate green innovation. *Economic Analysis and Policy, 83,* 60-79. https://doi.org/10.1016/j.eap.2024.06.010.

Wang, M., Xu, M., & Ma, S. (2021). The effect of the spatial heterogeneity of human capital structure on regional green total factor productivity. *Structural Change and Economic Dynamics, 59,* 427-441. https://doi.org/10.1016/j.strueco.2021.09.018.

Wu, H., Hao, Y., Ren, S., Yang, X., & Xie, G. (2021). Does internet development improve green total factor energy efficiency? Evidence from China. *Energy Policy, 153,* 112247. https://doi.org/10.1016/j.enpol.2021.112247.

Wu, J., Xia, Q., & Li, Z. (2022). Green innovation and enterprise green total factor productivity at a micro level: A perspective of technical distance. J*ournal of Cleaner Production, 344,* 131070. https://doi.org/10.1016/j.jclepro.2022.131070.

Wang, D., Liu, X., Hu, S., & Wu, S. (2025). The value of digital government transformation: Evidence from R&D subsidy efficiency in China. *International Review of Financial Analysis, 102,* 104106. https://doi.org/10.1016/j.irfa.2025.104106.

Wilson, C., & Mergel, I. (2022). Overcoming barriers to digital government: mapping the strategies of digital champions. *Government Information Quarterly, 39*(2), 101681. https:// doi.org/10.1016/j.giq.2022.101681.

Wang, H., & Guo, J. (2024). New way out of efficiency-equity dilemma: Digital technology empowerment for local government environmental governance. *Technological Forecasting and Social Change, 200,* 123184. https://doi.org/10.1016/j.techfore.2023.123184.

Xie, R., Fu, W., Yao, S., & Zhang, Q. (2021). Effects of financial agglomeration on green total factor productivity in Chinese cities: Insights from an empirical spatial Durbin model. *Energy Economics, 101,* 105449. https://doi.org/10.1016/j.eneco.2021.105449.

Xia, S. (2025). Digital government construction and civic policy participation: evidence from China. *Policy Sciences,* 1-23. https://doi.org/10.1007/s11077-025-09576-7.

Yang, C., Gu, M., & Albitar, K. (2024). Government in the digital age: Exploring the impact of digital transformation on governmental efficiency. *Technological Forecasting and*

Social Change, 208, 123722. https://doi.org/10.1016/j.techfore.2024.123722.

Yi, J. (2025). Digital government and carbon emissions: evidence from China. *Frontiers in Environmental Science, 13,* 1470724. https://doi.org/10.3389/fenvs.2025.1470724.

Yu, C., Gao, D., & Sun, H. (2025). The impact of digital government services on manufacturing productivity: China's pilot city insights. *Finance Research Letters, 73,* 106600. https://doi.org/10.1016/j.frl.2024.106600.

Yang, J. C., Chuang, H. C., & Kuan, C. M. (2020). Double machine learning with gradient boosting and its application to the Big N audit quality effect. *Journal of Econometrics, 216*(1), 268-283. https://doi.org10.1016/j.jeconom.2020.01.018.

Zhao, X., Nakonieczny, J., Jabeen, F., Shahzad, U., & Jia, W. (2022). Does green innovation induce green total factor productivity? Novel findings from Chinese city level data. *Technological Forecasting and Social Change, 185*, 122021. https://doi.org/10.1016/ j.techfore.2022.122021.

Zhang, L., & Zhang, X. (2025). Impact of digital government construction on the intelligent transformation of enterprises: Evidence from China. *Technological Forecasting and Social Change, 210,* 123787. https://doi.org/10.1016/j.techfore.2024.123787.

Zhang, P., Shihong, W. A. N. G., & Li, B. (2025). Harnessing government digital attention: Reducing carbon emissions through the pathways of digitalization. *Structural Change and Economic Dynamics*. https://doi.org/10.1016/j.strueco.2025.07.003.

Zhou, F., Wen, H., & Lee, C. C. (2022). Broadband infrastructure and export growth. *Telecommunications policy, 46*(5), 102347. https://doi.org/10.1016/j.telpol.2022. 102347.

Zhou, F., Chen, Y., & Wen, H. (2025). FinTech and Innovation: The Path to Upgrading Regional Carbon-Intensive Industries in China. *Emerging Markets Finance and Trade*, 1-18. https://doi.org/10.1080/1540496X.2025.2527762.

Appendix

Table A1 Definitions, measurements and sources of the main variables

Variable	Definition	Measurement	Source
GTFP	Green total factor productivity	Calculated by the Super-efficient SBM-Malmquist-Luenberger index	China Urban Statistical Yearbook
Bdmi-policy	Digital government construction	Pilot cities for the big data management institution reform	"Plan for Deepening the Reform of Party and State Institutions"
LnGDP	Economic development	The logarithm of urban GDP	China Urban Statistical Yearbook; The China Research Data Service Platform
Finance	Financial development	The proportion of the balance of various loans of financial institutions at the end of the year to GDP	
GOV	Government intervention	The proportion of local fiscal general budgetary expenditure to GDP	
Indus	Industrial structure	The proportion of the added value of the secondary industry in GDP	
Open	Opening-up	The proportion of the total volume of imports and exports to GDP	
HC	Human capital	The proportion of the number of college students in the local registered population	
DigAttention	Governmental digital attention	The proportion of the keyword frequency related to the governmental digital attention to the total keyword frequency	Government work reports
DigInfrus	Digital infrastructure construction	The proportion of the word frequency related to digital infrastructure in the government work report to the total word frequency of the text	Government work reports
DigTech	Digital technology innovation	The number of invention patents related to digital technology applied by enterprises within the urban jurisdiction in the current year	The National Intellectual Property Administration
DigEco	Digital economic development	Calculated based on the indicators of digital economy development	The China Research Data Service Platform